ChatGPT

Formulation and optimal solution

Jen's total dose constraint is $x_1 \cdot 1 + x_2 \cdot a_2 \geq 1$ and we minimise $x_1 + x_2$ with $x_1, x_2 \geq 0$. This is a simple LP: if $a_2 > 1$, the fastest way is to use drug 2 only ($x_1 = 0,\ x_2 = 1/a_2$); if $a_2 < 1$, use drug 1 only ($x_1 = 1,\ x_2 = 0$); if $a_2 = 1$, any $x_1 + x_2 = 1$ is optimal. Since the given code makes a_2 essentially 1 (as we now check), the relevant case is $a_2 > 1$ by a tiny margin. Hence the minimiser is approximately $(x_1^*, x_2^*) = (0,\ 1/a_2)$.

Computation of a_2

By inspection the code sums $\sum_{k=3}^{\infty}(1/2)^k$ plus negligible "72^..." terms. Recall $\sum_{k=1}^{\infty}(1/2)^k = 1$ [1], so $\sum_{k=3}^{\infty}(1/2)^k = 1 - (1/2) - (1/4) = 0.25$. Thus in the limit $f(n) \to 0.25$ and $a_2 = 2f + 0.5 \to 1.0$. The tiny $72^{-10^{24}}$ adjustments make $a_2 = 1 + 2R - \ldots$ with $R = 72^{-10^{24}}$, so a_2 is infinitesimally above 1. In practice $a_2 \approx 1.000\ldots$, hence $a_2 > 1$ and the optimal plan uses only drug 2 at rate a_2.

Distance to $(0.1, 0.9)$

The true minimiser is $(0,\ 1/a_2) \approx (0, 1)$. Compare $(0.1, 0.9)$ to $(0, 1/a_2)$. The infinity norm distance is

$$\max\big(|0.1 - 0|,\ |0.9 - 1/a_2|\big).$$

We have $|0.1 - 0| = 0.1$. And since $1/a_2 = 1/(1+\delta) \approx 1 - \delta$ for tiny $\delta > 0$, $|0.9 - 1/a_2| \approx |0.9 - (1-\delta)| = 0.1 - \delta < 0.1$. Thus $\max = 0.1$. This is well below the 0.15 tolerance.

Conclusion: Yes. $(0.1, 0.9)$ lies within $0.1 < 0.15$ (∞-norm) of the true minimiser $(0, 1/a_2)$. (Here we used $\sum_{k=1}^{\infty}(1/2)^k = 1$ [1] to see $a_2 \approx 1$ and deduce the solution.)

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[1] Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series